

Matrix Layer Rotation

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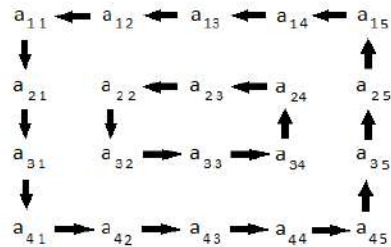
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You are given a 2D matrix of dimension $m \times n$ and a positive integer r . You have to rotate the matrix r times and print the resultant matrix. Rotation should be in anti-clockwise direction.

Rotation of a 4×5 matrix is represented by the following figure. Note that in one rotation, you have to shift elements by one step only.



Matrix Rotation

It is guaranteed that the minimum of m and n will be even.

As an example rotate the Start matrix by 2:

Start	First	Second
1 2 3 4	2 3 4 5	3 4 5 6
12 1 2 5 ->	1 2 3 6 ->	2 3 4 7

Author: [Heraldo](#)
 Difficulty: **Hard**
 Max Score: 80
 Submitted By: 79712

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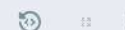


MORE DETAILS

- [Download problem statement](#)
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Change Theme Language Java 8



```
1 import java.util.*;
2
3 class Result {
4
5     public static void matrixRotation(List<List<Integer>> matrix, int r) {
6         int m = matrix.size();
7         int n = matrix.get(0).size();
8
9         int layers = Math.min(m, n) / 2;
10
11         for (int layer = 0; layer < layers; layer++) {
12
13             List<Integer> elements = new ArrayList<>();
14
15             int top = layer;
16             int bottom = m - 1 - layer;
17             int left = layer;
18             int right = n - 1 - layer;
19
20             // Top row: left -> right
21             for (int j = left; j <= right; j++) {
22                 elements.add(matrix.get(top).get(j));
23             }
24
25             // Right column: top+1 -> bottom
26             for (int i = top + 1; i <= bottom; i++) {
27                 elements.add(matrix.get(i).get(right));
28             }
29
30             // Bottom row: right-1 -> left
31             for (int j = right - 1; j >= left; j--) {
32                 elements.add(matrix.get(bottom).get(j));
33             }
34
35             // Left column: bottom-1 -> top+1
36             for (int i = bottom - 1; i > top; i--) {
37                 elements.add(matrix.get(i).get(left));
38             }
39         }
40     }
41 }
```

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Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

 Sample Test case 0 Sample Test case 1 Sample Test case 2 Sample Test case 3

Input (stdin)

Download

```
1 4 4 1
2 1 2 3 4
3 5 6 7 8
4 9 10 11 12
5 13 14 15 16
```

Your Output (stdout)

```
1 2 3 4 8
2 1 7 11 12
3 5 6 10 16
4 9 13 14 15
```